

	Deep Learning		Semester	7
Course Code	BCS714A		CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40		Total Marks	100
Credits	03		Exam Hours	03
Examination type (SEE)	Theory			
Course objectives: - Understand the basic concepts of deep learning. - Know the basic working model of Convolutional Neural Networks and RNN in decision making. - Illustrate the strength and weaknesses of many popular deep learning approaches. - Introduce major deep learning algorithms, the problem settings, and their applications to solve real world problems				
Teaching-Learning Process (General Instructions) These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes. - Lecturer method (L) need not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes. - Use of Video/Animation/Demonstration to explain functioning of various concepts. - Encourage collaborative (Group Learning) Learning in the class. - Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking. - Adopt Problem/Practical Based Learning (PBL), which fosters students' Analytical skills, develop design thinking skills, and practical skill such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it. - Use animations/videos to help the students to understand the concepts. - Demonstrate the concepts using PYTHON and its libraries wherever possible				
Module-1				
Introducing Deep Learning: Biological and Machine Vision: Biological Vision, Machine Vision: The Neocognitron, LeNet-5, The Traditional Machine Learning Approach, ImageNet and the ILSVRC, AlexNet, TensorFlow Playground. Human and Machine Language: Deep Learning for Natural Language Processing: Deep Learning Networks Learn Representations Automatically, Natural Language Processing, A Brief History of Deep Learning for NLP, Computational Representations of Language: One-Hot Representations of Words, Word Vectors, Word-Vector Arithmetic, word2viz, Localist Versus Distributed Representations, Elements of Natural Human Language.				
Text book 2 : Chapter 1, 2				
Module-2				
Regularization for Deep Learning: Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization and Under-Constrained Problems, Dataset Augmentation, Noise Robustness, Semi- Supervised Learning, Multi-Task Learning, Early Stopping, Parameter Tying and Parameter Sharing, Sparse Representations, Optimization for Training Deep Models: How Learning Differs from Pure Optimization, Basic Algorithms. Parameter Initialization Strategies, Algorithms with Adaptive Learning Rates.				
Text book 1 : Chapter 7 (7.1 to 7.10), Chapter 8 (8.1,8.3,8.4,8.5)				
Module-3				

	Convolution neural networks: The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data Types, Efficient Convolution Algorithms, Convolutional Networks and the History of Deep Learning. Text book 1 : Chapter 9 (9.1 to 9.8, 9.11)
	Module-4
	Sequence Modelling: Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks. Long short-term memory. Text book 1 : Chapter 10 (10.1 to 10.6, 10.10)
	Module-5
	Interactive Applications of Deep Learning: Natural Language Processing: Preprocessing Natural Language Data: Tokenization, Converting All Characters to Lowercase, Removing Stop Words and Punctuation, Stemming, Handling n-grams, Preprocessing the Full Corpus, Creating Word Embeddings with word2vec: The Essential Theory Behind word2vec, Evaluating Word Vectors, Running word2vec, Plotting Word Vectors, The Area under the ROC Curve: The Confusion Matrix, Calculating the ROC AUC Metric, Natural Language Classification with Familiar Networks: Loading the IMDb Film Reviews, Examining the IMDb Data, Standardizing the Length of the Reviews, Dense Network, Convolutional Networks, Networks Designed for Sequential Data: Recurrent Neural Networks, Long Short-Term Memory Units, Bidirectional LSTMs, Stacked Recurrent Models, Seq2seq and Attention, Transfer Learning in NLP. Text book 2 : Chapter-8
	Course outcomes (Course Skill Set): At the end of the course, the student will be able to: 1. Interpret the concepts of neural networks learning processes. 2. Illustrate deep learning methods using regularization and Optimization process 3. Design deep learning models using convolutional operations. 4. Analyze sequential data to build recurrent and recursive models. 5. Demonstrate the different interactive applications of deep learning.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**).

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:**Books**

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press, 2016.
https://www.deeplearningbook.org/lecture_slides.html
2. John Krohn, Grant Beyleveld, Aglae Bassens, Deep Learning Illustrated, A Visual, Interactive Guide to Artificial Intelligence, Pearson, 2022.

Web links and Video Lectures (e-Resources):

- <https://www.youtube.com/watch?v=VyWAvY2CF9c>
<https://www.youtube.com/watch?v=7sB052Pz0sQ>
https://www.youtube.com/watch?v=Mubj_fqiAv8
<https://www.coursera.org/learn/neural-networks-deep-learning>
- https://onlinecourses.nptel.ac.in/noc20_cs62/preview

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Programming Assignments, such as implementation of CNN and Recurrent neural network models - 10 Marks
- Group assignment (Group of two) on recent developments in Deep learning – Refer IEEE/ACM/Elsevier etc publications - 15 Marks

Software-Defined Networks		Semester	7
Course Code	BCM714B	CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	03
Examination type (SEE)	Theory		
Course objectives: ● To develop students’ ability to understand the concepts of traditional networks with its limitations and the need to move to Software Defined Networks. ● To develop the students’ ability to understand the fundamentals of SDN, its planar architecture and to understand the flexibility of multilevel pipeline processing. ● To provide the students with knowledge of the working of SDN between the controller and data plane and emphasis on the table matching. ● To create switches and designing networks by manually adding/deleting flow entries inside the table and learning to dissect the packets. ● To study industrial deployment use-cases of SDN			
Teaching-Learning Process (General Instructions) These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes. 1. Use of power point presentation 2. Think -pair and share techniques 3. Workshop on Mininet Network Emulator 4. Usage of OpenFlow 5. Brief about the real-world applications of SDN from the published papers 6. Think-Pair Share, Group Discussions, Peer Tutoring			
Module-1			
Foundations of Software Defined Networking: Basic Packet-Switching Terminology, The Modern Data Center, Traditional Switch Architecture, Autonomous and Dynamic Forwarding Tables, Evolution of Switches and Control Planes, The Evolution of Networking Technology, Software Defined Networking is Born. Textbook: Ch.1.1, 1.3, 1.4, 1.5, 2.1,3.1,3.3			
Module-2			
SDN Architecture and Protocols: Fundamental Characteristics of SDN, SDN Operation, SDN Devices, SDN Controller, SDN Applications, Alternate SDN Methods, OpenFlow Textbook: Ch. 4.1-4.6, 5.2			
Module-3			
SDN Controllers and Programming: Potential Drawbacks of Open SDN, Data Center Demands, Tunneling Technologies for the Data Center, Path Technologies in the Data Center, Ethernet Fabrics in the Data Center, SDN in Wide Area Networks, SDN in Campus Networks, SDN in Hospitality Networks. Textbook: Ch. 6.1, 7.2-7.5, 8.1, 8.3, 8.4			
Module-4			
SDN Applications and Real-World Use Cases: Reactive versus Proactive Applications, Analyzing Simple SDN Applications, A Simple Reactive Java Application, Using the Floodlight Controller, Creating Network Virtualization Tunnels, Traffic Engineering for Service Providers Textbook: Ch. 10.2-10.4,10.6,10.11,10.14			

	Module-5
	Challenges, Trends, and Advanced Topics: Open Source Licensing Issues, Profiles of SDN Open Source Users, OpenFlow Source Code, SDN Applications, Simulation, Testing, and Tools, OpenStack, Major SDN Acquisitions, SDN Startups. Textbook: Ch.11.2-11.4, 11.7, 11.9, 11.10, 12.8, 12.9
	Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Explain the challenges and opportunities associated with adopting SDN compared to traditional approaches to networking. 2. Identify the functions of the data plane and control plane 3. Analyze the performance of routing, optimizing traffic engineering using SDN. 4. Demonstrate the operations of SDN network consisting of switches and a centralized controller through simulation. 5. Demonstrate the industrial deployment use-cases of SDN.
	Assessment Details (both CIE and SEE) The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together. Continuous Internal Evaluation: • For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks. • The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered • Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned. • For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment. Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course. Semester-End Examination: Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (duration 03 hours). 1. The question paper will have ten questions. Each question is set for 20 marks. 2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), should have a mix of topics under that module. 3. The students have to answer 5 full questions, selecting one full question from each module. 4. Marks scored shall be proportionally reduced to 50 marks
	Suggested Learning Resources: Text Book Paul Goransson, Chuck Black, Timothy Culver, "Software Defined Networks: A Comprehensive Approach", 2nd Edition 2017.
	Reference Books

1. Nadeau, Thomas D., and Ken Gray. SDN: Software Defined Networks: an authoritative review of network programmability technologies. " O'Reilly Media, Inc.", 2013.
2. Coker, Oswald, and Siamak Azodolmolky. Software-defined Networking with OpenFlow: Deliver Innovative Business Solutions. Packt Publishing Ltd, 2017.

Web links and Video Lectures (e-Resources):

- <https://nptel.ac.in/courses/106106243>
- <https://www.opennetworking.org/> (Open Network Foundation)
- <https://www.opennetworking.org/wp-content/uploads/2014/10/openflow-spec-v1.3.0.pdf> (OpenFlow version 1.3)
- <http://mininet.org/> (Mininet Network Emulator).

Activity Based Learning (Suggested Activities in Class)/Practical-Based learning

- Course projects by the students (group of two) using Mininet Network Emulator and OpenFlow.

AD HOC AND SENSORS NETWORKS		Semester	7
Course Code	BCM714C	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	3
Examination nature (SEE)	Theory		
Course objectives: - To understand the concepts of sensor networks - To understand the MAC and transport protocols for ad hoc networks - To understand the security of sensor networks - To understand the applications of adhoc and sensor networks			
Teaching-Learning Process (General Instructions) These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes. - Lecturer method (L) needs not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes. - Use of Video/Animation to explain functioning of various concepts. - Encourage collaborative (Group Learning) Learning in the class. - Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking. - Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding. - Use any of these methods: Chalk and board, Active Learning, Case Studies.			
MODULE-1			
Introduction: The Communication Puzzle ,Applications of MANETs, Challenges. Routing in Ad Hoc Networks: Introduction , Topology-Based versus Position-Based Approaches Topology-Based Routing Protocols, Position-Based Routing, Other Routing Protocols. TB1: Ch 1: 1.11.2,1.3,1.4 Ch2: 2.1-2.5			
MODULE-2			
Broadcasting, Multicasting and Geocasting: Introduction, The Broadcast Storm, Multicasting, Geocasting. Wireless LANs: Introduction, Why Wireless LANs, Transmission Techniques, Medium Access Control Protocol Issues, The IEEE 802.11 Standard for Wireless LANs. TB1: Ch 3: 3.1-3.4, Ch 4: 4.1-4.5			
MODULE-3			
Wireless PANs: Introduction, Why Wireless PANs, The Bluetooth Technology, Enhancements to Bluetooth, The IEEE 802.15 Working Group for WPANs, Comparison between WPAN Systems, WLANs versus WPANs. TB1: Ch 5: 5.1-5.7			
MODULE-4			
Wireless Sensor Networks: Introduction, The Mica Mote, Sensing and Communication Range, Design Issues, Energy Consumption , Clustering of Sensors, Applications. TB1: Ch 8: 8.1-8.7			
MODULE-5			
Security: Distributed Systems Security, Security in Ad Hoc Networks, Key Management, Secure Routing, Cooperation in MANETs, Wireless Sensor Networks, Intrusion Detection Systems. TB1: Ch10: 10.1- 10.8			

Course outcomes (Course Skill Set):
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At the end of the course, the student will be able to:

- Explain the core concepts, applications, and challenges of Mobile Ad Hoc Networks (MANETs), including the communication puzzle.
- Apply Topology based and position-based routing protocols to address routing challenges in ad hoc networks.
- Evaluate the principles of Wireless LANs (IEEE 802.11) and Wireless PANs (Bluetooth, IEEE 802.15), comparing their transmission techniques and use cases.
- Identify design issues, energy-efficient strategies, and clustering techniques for Wireless Sensor Networks and their applications.
- Apply security mechanisms, including key management, secure routing, and intrusion detection, to ensure robust MANET and sensor network security.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.

The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered

Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.

For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

- Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (duration 03 hours).
- The question paper will have ten questions. Each question is set for 20 marks.
- There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), should have a mix of topics under that module.
- The students have to answer 5 full questions, selecting one full question from each module.
- Marks scored shall be proportionally reduced to 50 marks.

Suggested Learning Resources:

Books:

1. Ad Hoc and Sensor Networks – Theory and Applications, Carlos Corderio Dharma P. Aggarwal, World Scientific Publications, March 2006, ISBN – 981–256–681–3.

Reference Books:

- Wireless Sensor Networks: An Information Processing Approach, Feng Zhao, Leonidas Guibas, Elsevier Science, ISBN – 978-1-55860-914-3 (Morgan Kauffman).
- C. Siva Ram Murthy and B. S. Manoj, “Ad-hoc Wireless Networks”, Pearson Education, 2008.
- Erdal Cayirci and Chunming Rong, “Security in Wireless Ad Hoc and Sensor Networks”, John Wiley and Sons, 2009.

Activity-Based Learning (Suggested Activities in Class)/Practical-Based learning.

- **Course Project (Group of two)** :Examples - “Design of a Disaster Alert WSN System”, “Implementation of Bluetooth Mesh for Smart Campus” “Energy-aware Routing in Agricultural Sensor Networks”.

Mobile & Wireless Communication		Semester	07
Course Code	BCM714D	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40 hours Theory	Total Marks	100
Credits	03	Exam Hours	03
Examination nature (SEE)	Theory		
Course objectives: This course will enable to, - Understand the fundamentals of mobile communication systems. - Explain wireless transmission techniques, modulation and channel characteristics. - Illustrate wireless network protocols and architectures. - Analyze handoff strategies, routing and MAC protocols in mobile environments. - Explore emerging mobile technologies including 4G, 5G, and beyond.			
Teaching-Learning Process (General Instructions) These are sample Strategies that teachers can use to accelerate the attainment of the various course outcomes. - Lecturer method (L) need not to be only traditional lecture methods, but alternative effective teaching methods could be adopted to attain the outcomes. - Use of Video/Animation to explain functioning of various concepts. - Encourage collaborative (Group Learning) Learning in the class. - Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking. - Adopt Programming assignment, which fosters student’s Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.			
MODULE-1			
Introduction to Mobile and Wireless Communication: Cellular concept, comparison with wired communication, examples and applications. Wireless Transmission: Frequencies, signals, antennas, signal propagation, modulation techniques (ASK, FSK, PSK, QAM). Multiplexing and Multiple Access: FDMA, TDMA, CDMA, SDMA. Text Book 1: Rappaport: Ch. 1 (Intro), Ch. 2 (Wireless Transmission), Ch. 9 (Multiple Access) Text Book 2: Schiller: Ch. 2 (Wireless Transmission)			
MODULE-2			
Mobile Radio Propagation: Large-scale path loss models, Free space propagation, Reflection, Diffraction, and Scattering. Small-scale fading models, Doppler shift, delay spread, coherence bandwidth and time. Text Book 1: Rappaport: Ch. 3 (Propagation), Ch. 4 (Fading and Multipath), Ch. 5 (Modelling) Text Book 2: Schiller: Ch. 2 (Signal Propagation basics)			
MODULE-3			

Mobile Communication Systems: GSM Architecture, protocols, frame structure, call flow, services.
UMTS and CDMA2000: Features, architecture and working principles.

Text Book 2: Schiller: Ch. 4 (GSM), Ch. 5 (UMTS), Ch. 6 (CDMA2000)

Text Book 1: Rappaport: Ch. 10 (Wireless Systems)

MODULE-4

Wireless Networking: IEEE 802.11 WLAN, Bluetooth, ZigBee, WiMAX overview.
Mobile IP, DHCP, Routing in ad-hoc networks, MAC protocols.

Text Book 2: Schiller: Ch. 7 (Wireless LAN), Ch. 9 (Mobile IP), Ch. 10 (Ad-hoc)

Text Book 4: Stallings: Relevant chapters on WLAN, Bluetooth, WiMAX, and MAC

MODULE-5

Emerging Wireless Technologies: LTE, LTE-Advanced, 5G network architecture and applications.
Handoff strategies, QoS support in mobile communication, VOLTE, and IMS.

Text Book 3: Pahlavan: Ch. 11 (Emerging Networks)

Text Book 1: Rappaport: Ch. 13–14 (LTE/5G)

Text Book 4: Stallings: LTE, IMS architecture (final chapters)

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

- Explain the principles and technologies behind mobile and wireless communication systems.
- Apply modulation, transmission and multiplexing techniques for wireless systems.
- Analyze cellular architectures and propagation models for wireless environments.
- Evaluate wireless standards, network protocols and handoff mechanisms.
- Identify and describe emerging mobile technologies like 4G, LTE and 5G.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (duration 03 hours).

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), should have a mix of topics under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks.

Text Books:

1. Theodore S. Rappaport, 'Wireless Communications: Principles and Practice', 2nd Edition, Pearson.
2. Jochen Schiller, 'Mobile Communications', 2nd Edition, Pearson Education.
3. Kaveh Pahlavan and Prashant Krishnamurthy, 'Principles of Wireless Networks', Pearson.
4. William Stallings, 'Wireless Communication and Networks', Pearson Education.

Web links and Video Lectures (e-Resources):

NPTEL Wireless Communication (Prof. Ranjan Bose, IITD): <https://nptel.ac.in/courses/108102072>.
NPTEL Mobile Communication (Prof. S. Chakrabarti, IITKGP): <https://nptel.ac.in/courses/106105033>.
MIT OCW – Wireless Communication: <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-973-communication-system-design-spring-2006/>.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

Experiments - Simulation of wireless protocols using MATLAB or NS2/NS3. – 25 marks

Introduction to DBMS		Semester	7
Course Code	BCS755A	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	3
Examination nature (SEE)	Theory		
Course objectives: • To Provide a strong foundation in database concepts, technology, and practice. • To Practice SQL programming through a variety of database problems. • To Understand the relational database design principles. • To Demonstrate the use of concurrency in database. • To Design and build database applications for real world problems.			
Teaching-Learning Process (General Instructions) These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes. • Lecturer method (L) needs not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes. • Use of Video/Animation to explain functioning of various concepts. • Encourage collaborative (Group Learning) Learning in the class. • Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking. • Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding. • Use any of these methods: Chalk and board, Active Learning, Case Studies.			
MODULE-1			
Introduction to Databases: Introduction, Characteristics of database approach, Advantages of using the DBMS approach, History of database applications.			
Overview of Database Languages and Architectures: Data Models, Schemas, and Instances. Three schema architecture and data independence, database languages, and interfaces, The Database System environment.			
Textbook 1:Ch 1.1 to 1.8, 2.1 to 2.6			
MODULE-2			
Conceptual Data Modeling using Entities and Relationships: Entity types, Entity sets and structural constraints, Weak entity types, ER diagrams, Specialization and Generalization.			
Mapping Conceptual Design into a Logical Design: Relational Database Design using ER-to-Relational mapping			
Textbook 1: Ch 3.1 to 3.10, 9.1 & 9.2			
MODULE-3			
Relational Model: Relational Model Concepts, Relational Model Constraints and relational database schemas, Update operations, transactions, and dealing with constraint violations.			
Relational Algebra: Unary and Binary relational operations, additional relational operations (aggregate, grouping, etc.) Examples of Queries in relational algebra.			
Textbook 1: Ch 5.1 to 5.3, Ch 8.1 to 8.5			
MODULE-4			

SQL: SQL data definition and data types, Schema change statements in SQL, specifying constraints in SQL, retrieval queries in SQL, INSERT, DELETE, and UPDATE statements in SQL, Additional features of SQL

Normalization: Database Design Theory – Introduction to Normalization using Functional and Multivalued Dependencies: Informal design guidelines for relation schema, Functional Dependencies, Normal Forms based on Primary Keys, Second and Third Normal Forms, Boyce-Codd Normal Form, Multivalued Dependency and Fourth Normal Form, Join Dependencies and Fifth Normal Form.

Textbook 1: Ch 6.1 to 6.5, 14.1 to 14.7

MODULE-5

SQL: Advanced Queries: More complex SQL retrieval queries, Specifying constraints as assertions and action triggers, Views in SQL.

Concurrency Control in Databases: Two-phase locking techniques for Concurrency control, Concurrency control based on Timestamp ordering, Multiversion Concurrency control techniques, Validation Concurrency control techniques, Granularity of Data items and Multiple Granularity Locking.

Textbook 1: Ch 7.1 to 7.3, 21.1 to 21.5

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

- Demonstrate the basic elements of a database management system.
- Design entity relationship and convert entity relationship diagrams into RDBMS.
- Use Structured Query Language (SQL) for database manipulation.
- Apply normalization to increase the efficiency of database design.
- Illustrate the concepts of concurrency control techniques.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.

The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered

Any two assignment methods mentioned in the 220B2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.

For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

- Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (duration 03 hours).
- The question paper will have ten questions. Each question is set for 20 marks.
- There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), should have a mix of topics under that module.
- The students have to answer 5 full questions, selecting one full question from each module.
- Marks scored shall be proportionally reduced to 50 marks.

Suggested Learning Resources:

Text Books:

1. Fundamentals of Database Systems, Ramez Elmasri and Shamkant B. Navathe, 7th Edition, 2017, Pearson.

Reference Books:

1. Database management systems, Ramakrishnan, and Gehrke, 3rd Edition, 2014, McGraw Hill

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

Course Project (25 marks)

- For any problem selected
 - Develop the application having at least five tables & domain areas shall include health care, agriculture & so on.

Introduction to Algorithms		Semester	7
Course Code	BCS755B	CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	03
Examination type (SEE)	Theory		
Course objectives: • To learn the methods for analyzing algorithms and evaluating their performance. • To demonstrate the efficiency of algorithms using asymptotic notations. • To solve problems using various algorithm design methods, including brute force, greedy, divide and conquer, decrease and conquer, transform and conquer, dynamic programming, backtracking, and branch and bound. • To learn the concepts of P and NP complexity classes.			
Teaching-Learning Process (General Instructions) These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes. 1. Lecturer method (L) does not mean only the traditional lecture method, but different types of teaching methods may be adopted to achieve the outcomes. 2. Utilize video/animation films to illustrate the functioning of various concepts. 3. Promote collaborative learning (Group Learning) in the class. 4. Pose at least three HOT (Higher Order Thinking) questions in the class to stimulate critical thinking. 5. Incorporate Problem-Based Learning (PBL) to foster students' analytical skills and develop their ability to evaluate, generalize, and analyze information rather than merely recalling it. 6. Introduce topics through multiple representations. 7. Demonstrate various ways to solve the same problem and encourage students to devise their own creative solutions. 8. Discuss the real-world applications of every concept to enhance students' comprehension.			
Module-1			
INTRODUCTION: What is an Algorithm?, Fundamentals of Algorithmic Problem Solving, Important problem Types, Fundamental Data Structures, Analysis Framework, Asymptotic Notations and Basic Efficiency Classes, ,Analysis Framework, Asymptotic Notations and Basic Efficiency Classes,			
Chapter 1 (Sections 1.1 to 1.4), Chapter 2 (2.1, 2.2)			
Module-2			
FUNDAMENTALS OF THE ANALYSIS OF ALGORITHM EFFICIENCY: Mathematical Analysis of Non-recursive Algorithms, Mathematical Analysis of Recursive Algorithms.			
BRUTE FORCE APPROACHES: Selection Sort and Bubble Sort, Sequential Search and Brute Force String Matching.			
Chapter 2(Sections 2.3,2.4), Chapter 3(Section 3.1,3.2)			

	Module-3
	Exhaustive Search (Travelling Salesman problem and Knapsack Problem). Depth First search and Breadth First search. DECREASE-AND-CONQUER: Insertion Sort, Topological Sorting. DIVIDE AND CONQUER: Merge Sort, Binary Tree Traversals. Chapter 3(3.4,3.5), Chapter 4 (Sections 4.1,4.2), Chapter 5 (Section 5.1,5.3)
	Module-4
	TRANSFORM-AND-CONQUER: Balanced Search Trees (AVL Trees), Heaps and Heapsort. SPACE-TIME TRADEOFFS: Sorting by Counting: Comparison counting sort, Input Enhancement in String Matching: Horspool's Algorithm, Hashing. Chapter 6 (Sections 6.3,6.4), Chapter 7 (Sections 7.1,7.2, 7.3)
	Module-5
	DYNAMIC PROGRAMMING: Three basic examples, The Knapsack Problem and Memory Functions. THE GREEDY METHOD: Kruskal's Algorithm, Dijkstra's Algorithm, Huffman Trees and Codes. Chapter 8 (Sections 8.1,8.2), Chapter 9 (Sections 9.2,9.3,9.4)
Course outcome (Course Skill Set) At the end of the course, the student will be able to: 1. Explain the algorithm design steps and computational problem types. 2. Apply the asymptotic notational method to analyze the performance of the algorithms in terms of time complexity. 3. Demonstrate divide & conquer approaches and decrease & conquer approaches to solve computational problems. 4. Make use of the transform & conquer design approach to solve the given real-world or complex computational problems. 5. Apply greedy and dynamic programming methods to solve graph & string-based computational problems.	

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 220B2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**).

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:**Textbooks**

1. Introduction to the Design and Analysis of Algorithms, By Anany Levitin, 3rd Edition (Indian), 2017, Pearson.

Reference books

1. Computer Algorithms/C++, Ellis Horowitz, SatrajSahni and Rajasekaran, 2nd Edition, 2014, Universities Press.
2. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI.
3. Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education)

Web links and Video Lectures (e-Resources):

- Design and Analysis of Algorithms: <https://nptel.ac.in/courses/106/101/106101060/>

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

1. Problem Solving - Competitive programming (Hacker Rank/ Hacker Earth / Leetcode) – 10 Marks
2. Problem solving (Numerical examples) related to different algorithms – 15 Marks

	SOFTWARE ENGINEERING		Semester	7
	Course Code	BCS755C	CIE Marks	50
	Teaching Hours/Week (L:T:P:S)	3:0:0:0	SEE Marks	50
	Total Hours of Pedagogy	50	Total Marks	100
	Credits	04	Exam Hours	3
	Examination type (SEE)	Theory		
	Course objectives: To understand foundational principles and the evolving nature of software engineering. - To learn various software process models and their practical applications. - To acquire skills in gathering, modeling, and validating software requirements. - To apply Agile methodologies and understand core software engineering practices. - To build a foundation for software design, testing, and quality assurance.			
	Teaching-Learning Process These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes. 1. Lecturer method (L) needs not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes. 2. Use of Video/Animation to explain functioning of various concepts. 3. Encourage collaborative (Group Learning) Learning in the class. 4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking. 5. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it. 6. Introduce Topics in manifold representations. 7. Show the different ways to solve the same problem with different circuits/logic and encourage the students to come up with their own creative ways to solve them. 8. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding 9. Use any of these methods: Chalk and board, Active Learning, Case Studies			
	Module-1			
	Software and Software Engineering: The nature of Software, The unique nature of WebApps, Software Engineering, The software Process, Software Engineering Practice, Software Myths. Process Models: A generic process model, Process assessment and improvement, Prescriptive process models: Waterfall model, Incremental process models, Evolutionary process models, Concurrent models, Specialized process models. Unified Process , Personal and Team process models Textbook 1: Chapter 1: 1.1 to 1.6, Chapter 2: 2.1 to 2.5			
	Module-2			

	Understanding Requirements: Requirements Engineering, Establishing the ground work, Eliciting Requirements, Developing use cases, Building the requirements model, Negotiating Requirements, Validating Requirements. Requirements Modeling Scenarios, Information and Analysis classes: Requirement Analysis, Scenario based modeling, UML models that supplement the Use Case, Data modeling Concepts, Class-Based Modeling. Requirement Modeling Strategies : Flow oriented Modeling , Behavioral Modeling. Textbook 1: Chapter 5: 5.1 to 5.7, Chapter 6: 6.1 to 6.5, Chapter 7: 7.1 to 7.3
	Module-3
	Agile Development: What is Agility?, Agility and the cost of change. What is an agile Process?, Extreme Programming (XP), Other Agile Process Models, A tool set for Agile process . Principles that guide practice: Software Engineering Knowledge, Core principles, Principles that guide each framework activity. Textbook 1: Chapter 3: 3.1 to 3.6, Chapter 4: 4.1 to 4.3
	Module-4
	Software Design: Design within the context of software engineering, Design process and quality, Design concepts: abstraction, modularity, architecture, patterns. Architectural Design: Architectural styles and patterns, reference architectures, component-level design, designing class-based components, conducting component-level design, design for reuse. Textbook 1:Chapter 8: 8.1–8.6, Chapter 9: 9.1–9.5
	Module-5
	Software Testing: Introduction to software testing, Strategic approach, Test strategies for conventional and object-oriented software, Validation testing, System testing, White-box and Black-box testing, Basis Path Testing, Control structure testing. Software Quality: Concepts of quality, Software quality assurance, Reviews, Software reliability and metrics. Textbook 1: Chapter 14: Sections 14.1 to 14.5,Chapter 15: Sections 15.1 to 15.5, Chapter 19: Sections 19.1 to 19.5
Course outcome At the end of the course, the student will be able to : 1. Explain the software nature, engineering practices, myths, and software process models. 2. Apply requirements engineering, elicitation, modeling, and validation in software development. 3. Demonstrate agile principles, practices, and tools for software development agility. 4. Apply design concepts, process, and architecture for quality software development. 5. Explain software testing strategies and quality assurance for reliable software.	

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3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Textbook

Roger S. Pressman: Software Engineering – A Practitioner's Approach, 7th Edition, Tata McGraw Hill, 2010.

Web links and Video Lectures (e-Resources):

<https://www.geeksforgeeks.org/software-engineering/software-engineering/>

Activity-Based Learning (Suggested Activities in Class)/Practical-Based learning

- Course project (Group of two students): Simulation that covers all the phases of SDLC - 25 marks

INTRODUCTION TO COMPUTER NETWORKS		Semester	7
Course Code	BCM755D	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	03
Examination nature (SEE)	Theory		
Course objectives: This course will enable students to, • Study the TCP/IP protocol suite, switching criteria and Medium Access Control protocols for reliable and noisy channels. • Study network layer services and IP versions. • Discuss transport layer services and understand UDP and TCP protocols. • Demonstrate the working of different concepts of networking layers and protocols.			
Teaching-Learning Process (General Instructions) These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes. 1. Lecturer method (L) need not to be only traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes. 2. Use of Video/Animation to explain functioning of various concepts. 3. Encourage collaborative (Group Learning) Learning in the class. 4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking. 5. Adopt Problem Based Learning (PBL), which fosters student’s Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.			
MODULE-1			
Introduction: Data Communications, Networks, Network Types, Networks Models: Protocol Layering, TCP/IP Protocol suite, The OSI model, Introduction to Physical Layer: Transmission media, Guided Media, Unguided Media: Wireless. Switching: Packet Switching and its types. Textbook: Ch. 1.1 - 1.3, 2.1 - 2.3, 7.1 – 7.3, 8.3.			
MODULE-2			
Data Link Layer: Error Detection and Correction: Introduction, Block Coding, Cyclic Codes. Data link control: DLC Services: Framing, Flow Control, Error Control, Connectionless and Connection Oriented, Data link layer protocols, High Level Data Link Control. Media Access Control: Random Access, Controlled Access. Check Sum and Point to Point Protocol. Textbook: Ch. 10.1-10.4, 11.1 -11.4, 12.1 - 12.2			
MODULE-3			
Network Layer: Network layer Services, Packet Switching, IPv4 Address, IPv4 Datagram, IPv6 Datagram, Introduction to Routing Algorithms, Unicast Routing Protocols: DVR, LSR, PVR, Unicast Routing protocols: RIP, OSPF, BGP, Multicasting Routing-MOSPF. Textbook: Ch. 18.1, 18.2, 18.4, 22.2,20.1-20.3, 21.3.2			
MODULE-4			
Introduction to Transport Layer: Introduction, Transport-Layer Protocols: Introduction, User Datagram Protocol, Transmission Control Protocol: services, features, segments, TCP connections, flow control, Error control, Congestion control. Textbook: Ch. 23.1- 23.2, 24.1-24.3.4, 24.3.6-24.3.9			
MODULE-5			

Introduction to Application Layer: Introduction, Client-Server Programming, Standard Client-Server Protocols: World Wide Web and HTTP, FTP, Electronic Mail, Domain Name System (DNS), TELNET, Secure Shell (SSH).

Textbook: Ch. 25.1-25.2, 26.1-26.6

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

CO1 **Explain** the fundamentals of computer networks.

CO2 **Identify** key Internet applications and their protocols.

CO3 **Apply** the concepts of computer networks to demonstrate the working of various layers and protocols in communication network.

CO4 **Describe** the principles of protocol layering in modern communication systems.

CO5 **Compare** the classification of network services and architectures.

Suggested Learning Resources:

Textbook:

1. Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, Tata McGraw-Hill, 2013.

Reference Books:

1. Larry L. Peterson and Bruce S. Davie: Computer Networks – A Systems Approach, 4th Edition, Elsevier, 2019.
2. Nader F. Mir: Computer and Communication Networks, 2nd Edition, Pearson Education, 2015.
3. William Stallings, Data and Computer Communication 10th Edition, Pearson Education, Inc., 2014.

Web links and Video Lectures (e-Resources):

1. <https://www.digimat.in/nptel/courses/video/106105183/L01.html>
2. <http://www.digimat.in/nptel/courses/video/106105081/L25.html>
3. <https://nptel.ac.in/courses/10610>

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Simulation of networks and Protocols using open source simulation tools (E.g., NS3 or any relevant tools). - 25 marks.